

Lab #2 - Photodetectors

ECE 482/582, Fall 2003

Goals:

1. To gain a basic working knowledge of some common photodetectors that you will encounter in engineering applications.
2. To understand detector design trade-offs and limitations of detectors in terms of amplitude response and frequency response.
3. Be able to sketch circuit diagrams and the I-V behavior for the Si PIN photodiode in both photovoltaic and photoconductive modes.
4. Get some exposure to the electrical simulation and design aspects of using photodetectors in a system.

Reading:

Kuhn, *Laser Engineering*, pp. 458-466

Other Resources:

Horowitz and Hill, *The Art of Electronics*, 1989, Cambridge University Press

Alexander, Stephen, *Optical Communication Receiver Design*, Chapters 3 & 4

Moore, John, *Building Scientific Apparatus*, Section 4.8

Preparatory Questions:

1. Describe the basic physics of each of the following Photodetectors:
 - a. Photoconductor
 - b. PIN Photodiode
 - c. Avalanche Photodiode
2. Below is a diagram of the I-V characteristics of a photodiode and a photoconductor. It is important to understand the differences between them. As you can see the slope of the I-V curve (differential conductivity) for a reversed biased photodiode does not change with increased incident light intensity. However, the slope does change with light intensity for the photoconductor. If you are trying to detect a modulated light signal describe the best region of operation for the photodiode and photoconductor.

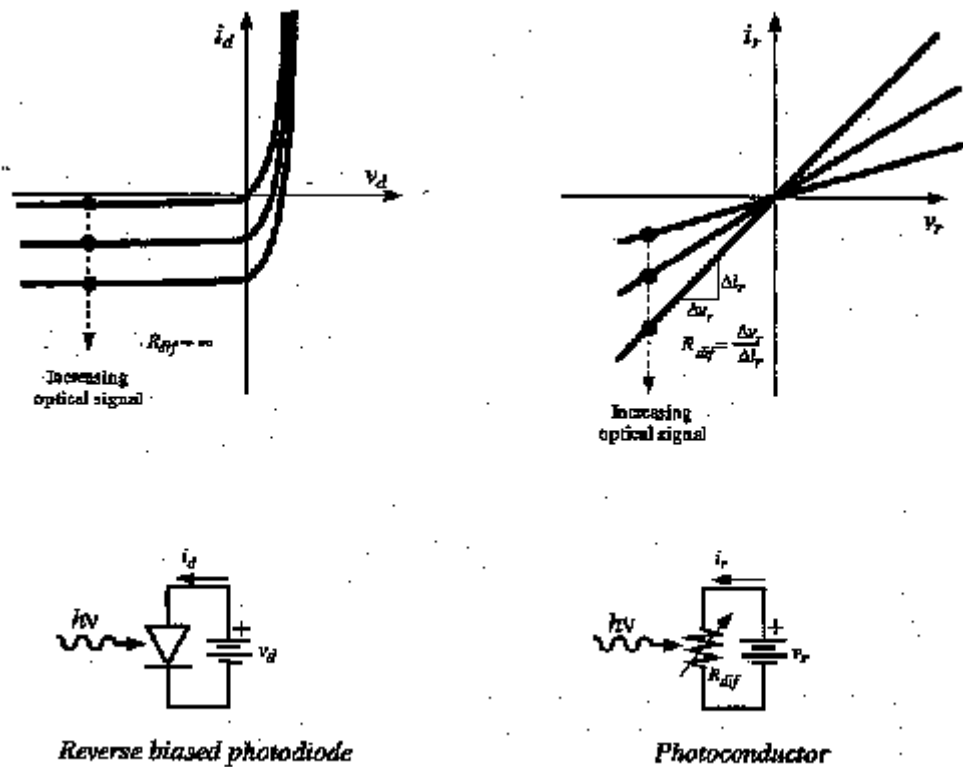
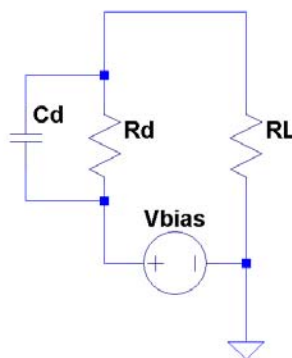
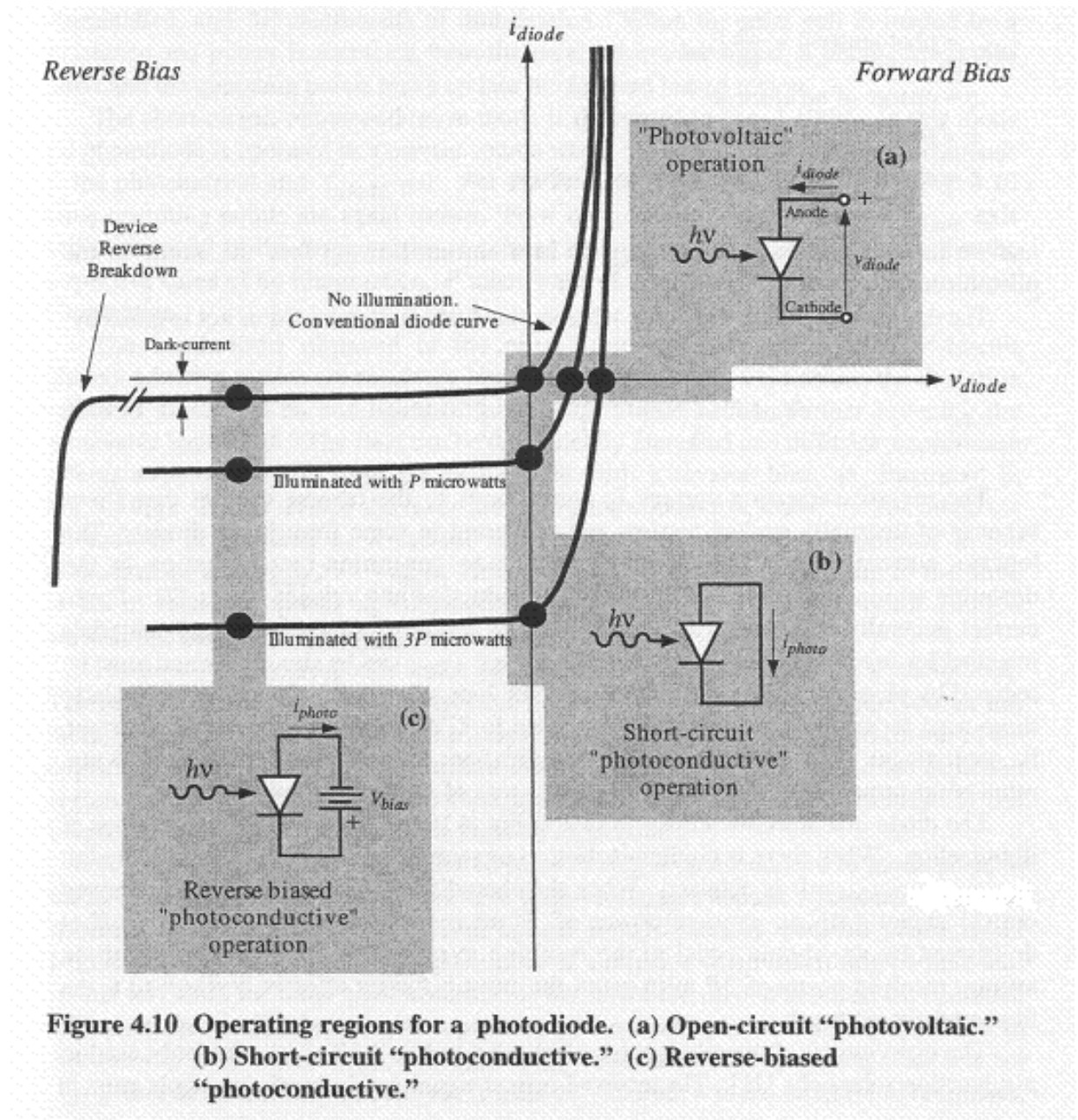


Figure 4.11 I-V characteristics of a photodiode and a photoconductor.
(Moore 1997)

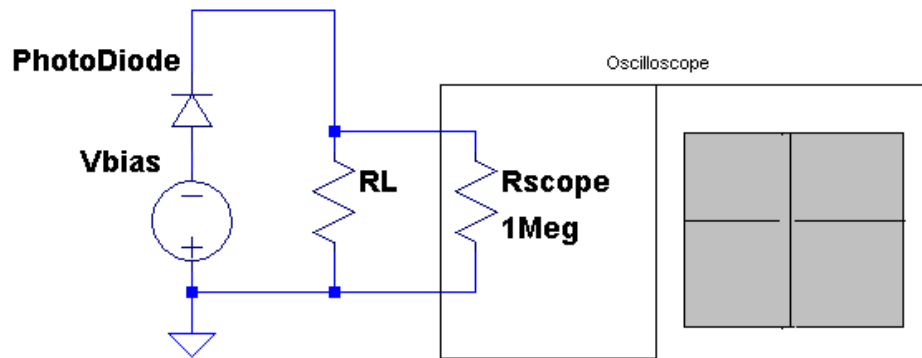
3. Below is a simple schematic of a basic biasing circuit commonly used for operating a photoconductor as a detector. An equivalent circuit model is used to represent the photoconductor where C_d is the device capacitance and R_d is the dark resistance. Find an analytical representation for the change in the output voltage (voltage across R_L) as a function of a change in resistance R_d . (It may help to redraw the circuit to make it look like something familiar...maybe a voltage divider?) When is this output voltage maximized? The bias voltage is selected to give the bias current that gives optimum detectivity. The device manufacturer will generally specify the optimum bias current.



4. The diagram below shows basic schematic representations of several photodiode detectors along with the V-I curve of a photodiode. Make sure you understand and memorize the three operation modes shown.



5. The following schematic shows a photodiode operated in one of the detector setups you will use in the lab. Which operational mode is this?



Based on the circuit diagram above and your knowledge of the I-V behavior of the PIN photodiode, answer qualitatively the following questions:

- How does V_{sig} change if the value of R_L that is in parallel to the 1Meg scope input resistor is increased? The combined resistance of the scope and R_L is called the effective load resistance.
- How does a detectors response time or “speed” change with a changing effective load resistance? What is the trade-off between amplitude response and speed?

Laboratory Experiment:

This lab examines two commonly used photodetectors: 1) the Si PIN photodiode and 2) the Si phototransistor. Keep in mind that it is very important to understand the Si PIN photodiode since you will use this in many subsequent labs and you will encounter it often in research and industry.

Characterization of a detector involves looking at the device response as a function of 1) the light intensity (**amplitude response**), 2) the wavelength of light (**spectral response**), and 3) a modulated input signal (**frequency response**). Today, we will look at the amplitude and frequency response for several detectors.

Procedure:

Part 1 – Amplitude and Frequency Response

When using voltage and current supplies disable the outputs before connecting a device to be tested. If the supply does not have an output disable feature then turn the supply voltage/current down to a minimum safe level before connecting/disconnecting any devices. After connecting the device increase the supply to the level desired.

1. Connect the -15V and $+15\text{V}$ supply to the LED modulation box supply leads.
2. Set the output of the function generator for a square wave of 10kHz and amplitude of 5.0V .
3. Connect the BNC connector for the LED modulation box to the function generator output. The LED should now be fairly bright and should be turning on and off $10,000$ times per second.

The detector box we will use contains two photodetectors, a Si PIN photodiode (with an active area dimension of $3\text{mm} \times 3\text{mm}$ and an integrated plastic lens in front of the diode to focus light onto the active area), and a Si Phototransistor ($1\text{mm} \times 1\text{mm}$ in dimension with no lens). The box also has three switches; one switch that changes between the two different photodetectors, another one to switch the load resistor values, and a third switch to switch in a 22V battery in series with the photodetector to reverse bias the detector (photoconductive mode).

4. Using the Si PIN photodiode (switch to Si PIN) in photoconductive mode (switch to PC) put the detector about 5mm from the LED source.
5. Connect the output BNC of the detector box to the oscilloscope. Sketch the detected signal seen on the oscilloscope. Repeat this for two or three load resistance values. *Be sure to label the voltage and time scales!
6. Measure and record the rise time (10% to 90%) of the signal for two load resistor settings. Remember that $t(\text{rise}) = 2.19 R_L C_{\text{Det}}$. Calculate the detector capacitance using your measurements.
7. Increase the spacing between the detector and the LED to 5cm . Repeat the measurements in steps 4 and 5.
8. Increase the spacing between the detector and the LED to 50cm . Repeat your measurements from steps 4 and 5. Is the signal still clearly visible at this distance? Recall that light intensity falls at $1/r^2$ as you move away from the source.
9. Move the detector switch to the photovoltaic (PV) position. Now the battery has been replaced by a short circuit. Repeat the same measurements you did for the photoconductive mode using two load resistor values.
10. Sketch an I-V diagram for both modes of a PIN photodiode. Draw a $-1/R_L$ “load line” on the I-V diagram for both a photoconductive and photovoltaic mode operation. Which mode will have the best signal linearity? Is this what you see in your measurements?
11. Now switch the detector to the Si phototransistor and place the PV/PC switch to the photoconductive (PC) mode. Again position the photodetector 5mm from the LED source.

12. Repeat the measurements made in steps 4 and 5 above. How does the response of this photodetector compare to the Si PIN photodiode? What is different?

Questions to Answer:

- Did you notice any saturation effects during your measurements? Describe the situation.
- How does the photovoltaic mode differ from the photoconductive mode?
- Do your measurements and observations indicate that the frequency response is bandwidth-limited by the source, detector, and/or the oscilloscope? Why is this important? When you measured the signal rise time can you expect the measured value to be accurate? Is there a better way to measure the rise time in order to prevent limitations of the source?
- How is the amplitude response affected by the load resistor value? Does this make sense according to Ohm's Law?
- According to your observations is the frequency response better for the PIN photodiode in photovoltaic mode or photoconductive mode? In other words, which mode is best for high-speed operation?

Part 2 – Design Principles

It is also very important to take into consideration the system and application you intend for the photodetector. You have learned basic ways to observe the amplitude and frequency response of a detector in a basic circuit configuration. Now let's look at how we can predict a photodetector's response in a system and learn some design rules to use in practice.

We will be using a circuit simulation tool called LTSPICE for this section. Since not all of you have previous experience using a SPICE simulator, you will work with a circuit that is pre-created and saved on the lab computer. OK. Let's have some fun!

1. Open up the SPICE simulator tool by clicking on the icon labeled "swCAD III". Go to the file menu and open the circuit file labeled "Photodiode". A circuit schematic should open up in the work window.
2. The opened Photodiode circuit is an "equivalent" circuit representation of an actual PIN photodetector that is used in a 10Gbit/s optical communications system. Notice that there are no diode symbols in the schematic. That is because everything is represented in terms of basic electrical building blocks (resistance, capacitance and inductance). I_1 (current source 1) represents the optically generated photocurrent in the depletion region of the photodiode. R_j is the junction resistance and represents the differential resistance of the diode. For a high-quality diode operating in the reversed bias mode this resistance is typically greater than 10^6 ohms and can be ignored. C_j is the junction capacitance and arises from the separation of charge in the depletion region. The series resistance R_s is due to bulk contact resistance (the contact of the diode to the outside world). C_b is the bond pad capacitance. L_s is the series inductance due to a wire contacting cathode of the diode to other circuit components in the system. C_p is the package or "parasitic" capacitance due to the packaging of the detector. Draw this circuit in your notebooks and label the values.

3. We will simulate the circuit with both a “transient analysis” and an “AC analysis”. Click on the little running man button on the toolbar. The program will ask you to select an analysis to run. Choose the “**.ac lin 200 100meg 20G**” line and click OK. Select the V(out) waveform to plot. The program will generate a plot of the AC frequency response of the detector circuit. The solid line is the magnitude and the dotted line is the phase. What is the -3dB cutoff frequency? Record this in your notebook.
4. Play around with the component values and discover what is limiting the speed of this detector.
5. Now let's get a different perspective. Close the entire program (without saving!) and re-start it. Again open the “Photodiode” circuit model. Click on the run button and this time choose the “**.tran 0 200p 0**” line. Choose both (hold down the Ctrl key while selecting) the V(out) and I(I1) waveforms to plot. The program will generate a plot of the input current pulse and the simulated output voltage pulse. Draw or print the plots and record them into your notebook.
6. Change component values and discover the effect on V(out). Record your findings.
7. You may recall that the Fourier transform of the impulse response of a linear system gives the frequency response. Let's try it! While the plot window is active choose the view menu and select FFT. Select the V(out) waveform only and click OK. You will need to adjust the scales on both axes by clicking on them and typing in values. Turn off the logarithmic scale for the horizontal axis. Set the vertical axis to a range of no more than 20dB. Set the horizontal axis to go from 100MHz to 15GHz. How does this frequency response plot compare to the one you generated from the AC analysis?
8. Close the program without saving.

Questions to Answer:

- What was the major speed limiting electrical property of the photodetector?
- A general design rule of thumb for common bond wires is that 1mm of bond wire contributes 1nH of inductance. How did the bond wire inductance affect the response of the photodetector? Calculate how long the bond wire is for the design (before you adjusted values) of our detector that operates in a 10GHz system. What can you say about bond wire length for higher frequencies?
- Explain what you saw when you increased the package capacitance? What does this mean?

Conclusion:

You should now be familiar with several different photodetectors and understand the basic design rules for working with them in real applications. Take some time to think about and summarize the concepts that you discovered so they will be useful to you in your other classes and future experimental and design work.